# Assessment Plan for PHY4605 (Semester 1 2026/2027)

**Course code and name:** Kaedah Pengkomputeran dalam Fizik / Computational Methods in Physics (PHY4605)

**Course type:** ~~1. General / Basic~~ ~~2. Fully Practical~~ 3. Focused

**Credits:** 3 (2+1)

**Total Student Learning Time:** 120 hours

**Cohort:** ~~Year 1~~ ~~Year 2~~ Year 3 Year 4

**Lecturer:** Dr Muhammad Khairul Adib Muhammad Yusof

| Course type | Cognitive | Psychomotor | Affective |
|---|---|---|---|
| General course | 60–80% | 5–40% | 5–40% |
| Focused course | 60–80% | 5–40% | 5–40% |
| Practical / Fully Practical course | 30–60% | 30–60% | ≥5% |

| Activity | Type of Assessment | Week | Programme Outcome Contribution Course-weight allocation (%) | | | | Total (%) | Remarks |
|---|---|---|---|---|---|---|---|---|
| | | | PO1 | PO2 | PO8 | PO9 | | |
| **Practical portfolio** | Summative AI-assisted | 2–11 | | 15 | | | 15 | Ten packets; best 8 counted; 12% group + 3% individual defence |
| **Individual lab test** | Summative AI-assisted | TBC | | 10 | | | 10 | Scaffolded verification + parameter/defect variant + defence |
| **Capstone investigation** | Summative AI-assisted | 13–14 | | | 5 | 10 | 15 | Bounded group; checkpoints 06–12; presentation, defence, handoff |
| **Test 1** | Summative AI-free | 6 | 10 | | | | 10 | 90 min open-book AI-free; Weeks 01–05 |
| **Test 2** | Summative AI-free | 12 | 10 | | | | 10 | 90 min open-book AI-free; Weeks 06–11 |
| **Final examination** | Summative AI-free | 16–17 | 40 | | | | 40 | 120 min cumulative open-book AI-free |
| **Grand total** | | | **60** | **25** | **5** | **10** | **100** | |

*Notes.* Week 01 is formative onboarding and is not one of the ten counted portfolio packets. The exact week for the individual lab test is not fixed in the current assessment blueprint and remains lecturer-confirmed. The percentages in the PO columns allocate each assessment's existing course weight to its principal evidence; each assessment is assigned to one or two principal POs only, and these are not additional marks.

# Mapping CO–PO for PHY4605

| CO | Course Outcome (CO) | Programme Outcome (PO) | | | |
|---|---|---|---|---|---|
| | | PO1<br>C | PO2<br>PP | PO8<br>KK | PO9<br>LS |
| 1 | **Analyse** a familiar physics model and numerical evidence by identifying assumptions, variables, units, expected behaviour, and appropriate validation checks.<br>(C4–C5) | ✓ | | | |
| 2 | **Apply and adapt** scaffolded numerical algorithms and MATLAB code to produce reproducible evidence, diagnose clear defects, and interpret the physical result.<br>(P4–P5) | | ✓ | | |
| 3 | **Communicate and defend** a bounded computational investigation, including limitations, responsible AI decisions, reproducibility, and individual/team accountability.<br>(A4) | | | ✓ | ✓ |
| | **Course-weight contribution (%)** | **60** | **25** | **5** | **10** |

**Programme outcome codes**

PO1 – Knowledge (C)
PO2 – Technical / practical / psychomotor skills (PP)
PO3 – Critical and scientific thinking (CTPS)
PO4 – Communication skills (CS)
PO5 – Social and responsibility skills (TS)
PO6 – Professionalism, values, ethics and morality (EM)
PO7 – Lifelong learning and information management (LL)
PO8 – Personal and entrepreneurship skills (KK)
PO9 – Leadership, autonomy and responsibility (LS)
PO10 – Numeracy skills (NS)

*Mapping rule.* A checkmark indicates direct evidence in the current PHY4605 assessment blueprint. The selected PO columns and allocations follow the evidence emphasis of the new assessment system: written assessments primarily evidence knowledge and analysis (PO1), practical verification evidences technical skills (PO2), and the capstone evidences personal/entrepreneurship and leadership/autonomy responsibilities (PO8 and PO9). Each assessment is assigned to one or two principal POs only. This proposed operational mapping should be checked against the programme's currently approved PO definitions before formal submission.